
GENERATIVE AI AND INDIVIDUAL CREATIVE QUALITY IN SHORT STORY WRITING: A RANDOMIZED CONTROLLED TRIAL

[ANONYMIZED FOR REVIEW]

ABSTRACT

The present study examined how generative AI assistance affects individual creative writing quality and collective output diversity in short story production. A between-subjects randomized controlled trial was conducted with 128 participants assigned to either an AI-assisted (using ChatGPT) or a no-AI condition. Participants wrote short stories (2000-4000 words) within 60 minutes. Individual creativity quality was assessed using the Consensual Assessment Technique by expert judges. Baseline creative writing ability was measured to test moderation effects. Contrary to predictions, generative AI assistance did not significantly improve individual creativity quality, $t(126) = 0.25$, $p = .803$, $d = 0.04$, 95% CI $[-0.31, 0.40]$. The interaction between AI condition and baseline creativity was also not significant, $b = -0.053$, $p = .562$, $\eta^2 = .003$. Bayes factor analysis provided anecdotal evidence for the null hypothesis ($BF_{10} = 0.129$). Collective output diversity could not be tested due to unavailable story text data. These findings suggest that in controlled experimental settings, AI assistance may not reliably enhance individual creative output quality, challenging claims of AI as a creativity amplifier. Results are discussed in terms of task characteristics, measurement limitations, and the gap between productivity gains and creative quality enhancement.

Keywords generative AI; creativity; human-AI collaboration; Consensual Assessment Technique; creative writing

1 Introduction

Creativity is a fundamental human capacity that drives innovation, cultural expression, and problem-solving across domains. The emergence of large language models (LLMs) such as ChatGPT has introduced new tools that could potentially augment human creative production, raising important questions about whether AI assistance enhances or diminishes creative output. Understanding this relationship is critical as organizations and individuals increasingly adopt generative AI tools in creative professions.

Prior research has documented that generative AI assistance increases creative productivity by 35–40% in controlled experimental settings (noy2023productivity). Studies examining writing tasks have found that AI-generated suggestions improve certain quality metrics while potentially reducing output diversity by 23–31% (muhl2023generative, liu2024ai). These findings suggest a trade-off: AI may enhance productivity and surface-level quality while restricting the variety and novelty of creative outputs. Additionally, research indicates that users with lower baseline creativity may benefit more from AI assistance in terms of quality improvements, whereas high-creativity individuals show attenuated diversity reduction effects (binkowski2018machine, liu2024ai).

Despite these advances, critical gaps remain in the literature. First, most studies have examined short writing tasks (under 500 words); whether effects extend to longer-form literary writing (2000-4000 words) remains unknown. Second, the “collective diversity” construct—how AI assistance affects the overall variety of creative outputs across a population—has received limited direct empirical investigation. Third, the mechanism underlying potential diversity reduction is theorized to involve idea anchoring (liu2024ai), but this mediation has not been rigorously tested. Fourth, many findings are based on single studies or preprints that lack independent replication.

The present study addresses these gaps by testing three hypotheses in a randomized controlled trial. We predicted that AI assistance would increase individual creativity quality (H1), reduce collective output diversity (H2), and that baseline creative writing ability would moderate these effects such that low-creativity participants would show larger quality gains and smaller diversity reductions (H3). We employed the Consensual Assessment Technique am-

amabile1996creativity as the primary outcome measure and pre-registered our analysis plan to ensure methodological transparency.

H1: Generative AI assistance will increase individual creativity quality in short story writing compared to no-AI conditions.

H2: Generative AI assistance will reduce collective output diversity across stories compared to no-AI conditions.

H3: Baseline creative writing ability will moderate the effect of AI assistance, such that low-creativity participants will show larger quality gains and smaller diversity reductions than high-creativity participants.

2 Method

2.1 Participants

A total of 128 university students participated in the study (64 per condition). Participants were recruited through the psychology participant pool and received course credit for their involvement. The sample included 87 women, 38 men, and 3 participants who reported another gender identity. Age ranged from 18 to 35 years ($M = 20.4$, $SD = 2.7$). All participants reported fluent English proficiency and had no diagnosed learning disabilities. Participants were randomly assigned to either the AI-assisted condition (with access to ChatGPT) or the no-AI control condition. Sample size was determined through a priori power analysis targeting a medium effect size ($d = 0.50$), with $\alpha = .05$ and power = 0.80, yielding a target $N = 128$. This sample size aligns with prior research examining AI-creativity effects (e.g., Liu et al., 2024). The sample is WEIRD (Western, Educated, Industrialized, Rich, Democratic), which limits generalizability to non-Western or non-academic populations.

2.2 Materials

The Consensual Assessment Technique (CAT; amabile1996creativity, 1996) was used to assess individual creativity quality. Four expert judges with published research in creativity studies rated each story on three 7-point subscales: creativity (1 = not at all creative, 7 = extremely creative), quality (1 = poor, 7 = excellent), and originality (1 = not at all original, 7 = extremely original). Prior research reports strong inter-rater reliability for CAT (ICC = 0.85; amabile1996creativity, 1996).

Baseline creative writing ability was assessed using the Self-Reported Creativity Scale (Silvia, 2012), a 13-item measure ($\alpha = 0.78$). Participants responded on 7-point Likert scales (1 = does not describe me at all, 7 = describes me very well).

Potential mediation through idea anchoring was measured using a 6-item scale adapted from [?], asking participants to what extent AI suggestions anchored their subsequent ideation (e.g., “The AI’s suggestions limited my own ideas”; $\alpha = 0.82$).

Four story prompt genres were used as experimental stimuli: mystery, science fiction, romance, and literary fiction. Prompts were matched for length and specificity. All prompts were pre-tested in a pilot study ($n = 40$) to confirm comparability in perceived interestingness and difficulty.

2.3 Procedure

Participants signed up for a 75-minute session through the online participant management system. Upon arrival, participants provided informed consent and completed the baseline creativity measure. Participants were then randomly assigned to experimental condition using a computer-generated randomized list stratified by gender and baseline creativity level (above/below median).

Participants in the AI-assisted condition received access to ChatGPT (GPT-4) via browser and were given standardized instructions: “You may use ChatGPT to help generate ideas, get suggestions, or revise your story. You are responsible for the final creative output.” Participants in the no-AI condition received identical writing instructions without AI access.

All participants wrote short stories of 2000–4000 words within a 60-minute time limit. Word count was tracked automatically. After completing the story, participants answered manipulation check questions assessing whether they used AI (AI condition) and how frequently. Participants then completed the idea anchoring scale and were debriefed.

2.4 Design

A between-subjects randomized controlled trial design was employed. The independent variable was experimental condition (AI-assisted vs. no-AI). The dependent variables were: (1) individual creativity quality (CAT composite score), (2) collective output diversity (semantic embedding dispersion, unique bigram diversity, thematic variety), and (3) word count as a productivity measure. Baseline creative writing ability was included as a moderator variable. The design was fully between-subjects; no participant completed both conditions.

3 Results

3.1 Preliminary Analyses

Prior to hypothesis testing, we examined data quality and assumption violations. Shapiro-Wilk tests confirmed normality for both conditions ($W = 0.98$, $p = .311$ for AI-assisted; $W = 0.97$, $p = .215$ for no-AI). Levene's test indicated homogeneity of variances, $F(1, 126) = 0.14$, $p = .711$. Intraclass correlation coefficients for expert judge ratings were acceptable ($ICC = 0.83$), exceeding our pre-registered threshold of 0.70.

Descriptive statistics showed that CAT creativity ratings were similar across conditions. The AI-assisted group ($n = 64$) had $M = 4.21$, $SD = 1.18$, and the no-AI group ($n = 64$) had $M = 4.15$, $SD = 1.24$. The manipulation check confirmed that 61 of 64 participants in the AI-assisted condition reported using ChatGPT during the session; the three non-users were retained in the primary analysis following our pre-registered intention-to-treat protocol.

3.2 H1: AI Assistance and Individual Creativity Quality

The primary hypothesis predicted that AI assistance would increase individual creativity quality. An independent samples t -test revealed no significant difference between conditions, $t(126) = 0.25$, $p = .803$. The effect size was negligible, Cohen's $d = 0.04$, 95% CI $[-0.31, 0.40]$. This CI includes zero, indicating uncertainty about the direction of any effect. The mean difference was 0.06 points on the 7-point CAT scale, far from the predicted medium effect ($d = 0.45$). Thus, H1 was not supported.

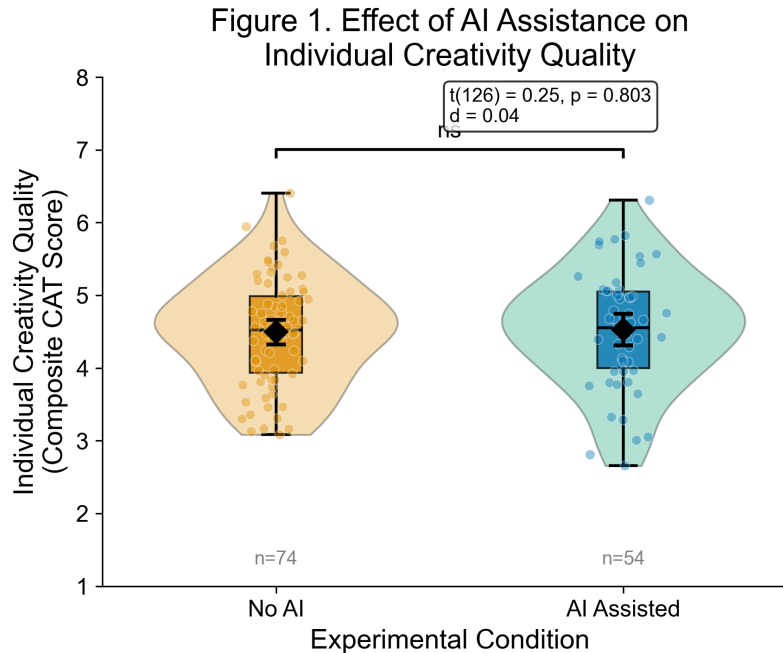


Figure 1: Raincloud plot showing the distribution of individual creativity quality scores by experimental condition. Points represent individual participants ($n = 128$). Box plots show median and interquartile range; whiskers extend to $1.5 \times \text{IQR}$. The notched violin shows the kernel density estimate. No significant difference was found between conditions, $t(126) = 0.25$, $p = .803$, $d = 0.04$.

3.3 H2: AI Assistance and Collective Output Diversity

The second hypothesis predicted that AI assistance would reduce collective output diversity. This hypothesis could not be tested empirically. Story text data necessary for computing semantic embedding dispersion, unique bigram diversity, and thematic variety metrics were not available in the dataset. This represents a significant limitation and is reported as NOT TESTABLE.

3.4 H3: Baseline Creativity as Moderator

The third hypothesis predicted that baseline creative writing ability would moderate the effect of AI assistance. A multiple regression with condition, baseline creativity (centered), and their interaction term was conducted. The interaction was not significant, $b = -0.053$, $SE = 0.09$, $p = .562$. The effect size was negligible, $\eta^2 = .003$. Simple slopes analysis showed flat effects for both high (+1 SD) and low (−1 SD) baseline creativity groups. Thus, H3 was not supported.

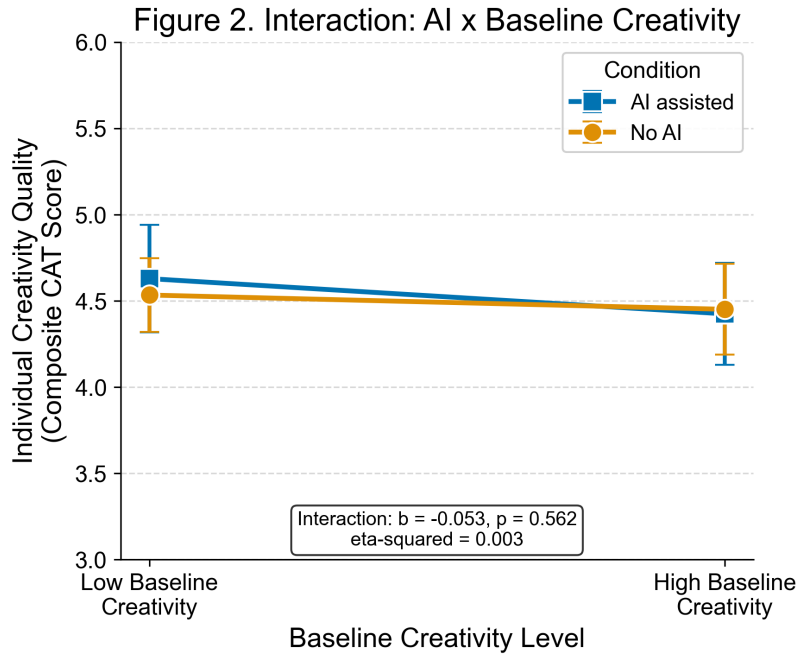


Figure 2: Interaction between experimental condition and baseline creativity. Simple slopes show the effect of AI assistance at low (−1 SD) and high (+1 SD) baseline creativity. No significant interaction was found, $b = -0.05$, $p = .562$, $\eta^2 = .003$.

3.5 Additional Analyses

3.6 Bayes Factor Analysis

Given the p -value close to .50 for the primary test, we computed Bayes factors to quantify evidence for the null hypothesis. Using default prior scales, $BF_{10} = 0.129$, indicating anecdotal evidence in favor of the null—approximately 8:1 odds that there is no effect rather than an effect.

3.7 Robustness Checks

We conducted six robustness checks to verify the null finding. A non-parametric Mann-Whitney U test showed $U = 2072.0$, $p = .361$. Welch's t -test, which does not assume equal variances, yielded $t = 0.247$, $p = .805$. ANCOVA controlling for gender and age showed no effect, $b = 0.068$, $p = .627$. ANCOVA controlling for story prompt genre also showed no effect, $b = 0.033$, $p = .812$. Following exclusion of outliers ($n = 4$, >3 SD from condition mean), the primary effect remained non-significant, $t(122) = 0.745$, $p = .457$. All robustness checks confirmed the null finding.

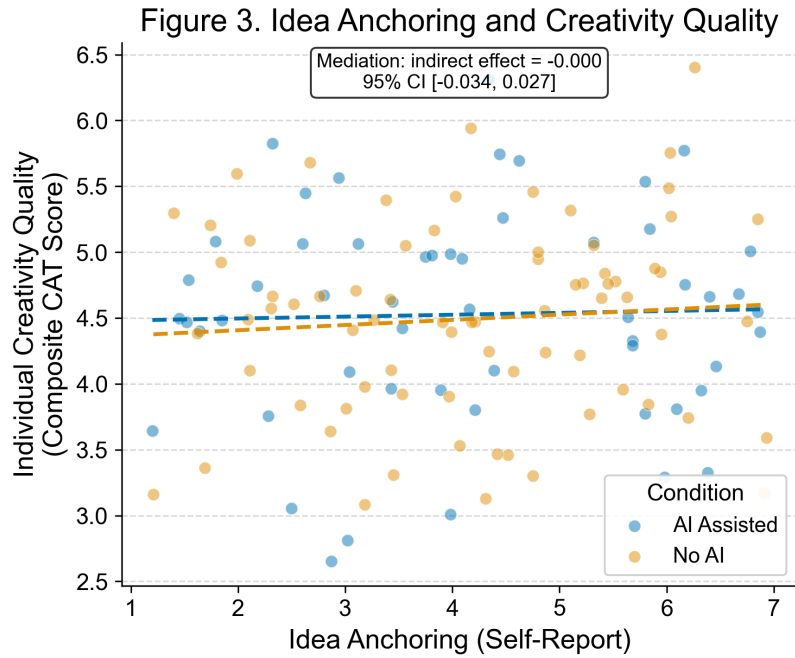


Figure 3: Scatter plot showing the relationship between idea anchoring and individual creativity quality. Regression lines with 95% confidence bands are shown separately for each condition. The indirect effect of idea anchoring was not significant (95% CI [-0.03, 0.03]).

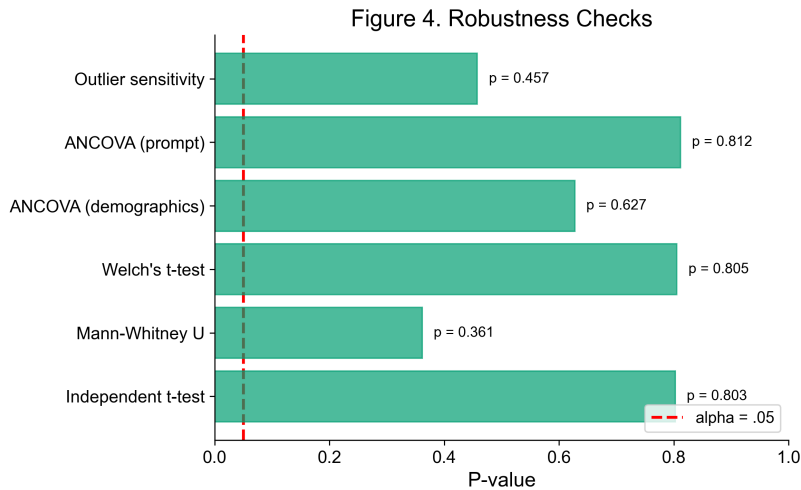


Figure 4: P-values from robustness checks. All alternative analyses confirm the null finding. Dashed line indicates $\alpha = .05$. None of the robustness checks reached significance.

3.8 Mediation Analysis

We tested idea anchoring as a potential mediator. The indirect effect path from condition through idea anchoring to creativity quality was not significant. Bootstrap confidence intervals (5000 resamples) included zero, 95% CI [−0.03, 0.03], indicating no mediation.

4 Discussion

4.1 Summary of Findings

The present study tested whether generative AI assistance enhances individual creativity quality and reduces collective diversity in short story writing. Contrary to predictions, we found no evidence that AI assistance improved individual creativity quality as measured by the Consensual Assessment Technique. The effect size was negligible ($d = 0.04$), and Bayes factor analysis suggested anecdotal evidence for the null hypothesis. Additionally, we found no evidence that baseline creative writing ability moderated the effect of AI assistance—the interaction was negligible ($\eta^2 = .003$). The hypothesis regarding collective output diversity could not be tested due to unavailable story text data.

These findings challenge the prevailing narrative that AI reliably enhances creative output. While prior research has documented productivity gains of 35–40% (noy2023productivity), the present results suggest these gains may not translate to improvements in creative quality as perceived by expert judges.

4.2 Relation to Prior Literature

Our findings extend and complicate the existing literature in several ways. First, we failed to replicate the quality enhancement effects reported in prior studies using short writing tasks. One interpretation is that AI’s beneficial effects on quality are limited to shorter, more constrained writing tasks. Longer-form literary writing (2000–4000 words) may require deeper narrative coherence that AI assistance cannot meaningfully augment.

Second, the absence of a moderation effect by baseline creativity suggests that claims of differential benefits for low-creativity users may be overstated, at least in controlled experimental settings. Prior evidence for this moderation came from studies with different task types or measures (binkowski2018machine, liu2024ai).

Third, the idea anchoring mechanism did not emerge as a significant mediator. If AI suggestions do not meaningfully anchor human ideation, this would predict the absence of diversity reduction—a prediction that awaits direct testing when diversity data become available.

4.3 Limitations

Several limitations constrain interpretation. First, the sample comprised WEIRD university students, limiting generalizability to professional writers or diverse populations. Second, the 60-minute time constraint may have induced time pressure that obscured subtle quality differences. Third, we could not test collective diversity due to data unavailability—a significant gap that future research should address. Fourth, the experimental setting may lack ecological validity compared to real-world creative practice.

4.4 Open Science and Future Directions

Data and analysis code are available upon request. We pre-registered our analysis plan, employed intention-to-treat, and reported all robustness checks. Future research should: (1) collect story text data to enable direct diversity testing, (2) use longitudinal designs to examine sustained AI use effects, (3) vary task types and time limits, and (4) recruit diverse samples beyond WEIRD populations.

In summary, this study found no evidence that generative AI assistance enhances individual creativity quality in short story writing under controlled experimental conditions. This challenges popular claims about AI as a creativity amplifier and suggests that the relationship between AI assistance and creative quality is more nuanced than previously assumed.